

AI Applications in Finance & Banking

1. AI & Machine Learning in Consumer Finance

Machine learning models now operate invisibly in the background of everyday financial services, influencing prices, loan rates, and user experiences in real time.

Dynamic Pricing & Demand Manipulation

- ML models work in the background even as you book flights, adjusting prices based on your browsing behavior.
- Models hike prices based on the order and frequency of searches — cookies track demand spikes and allow real-time price manipulation based on search volume.
- Loan interest rates can be spiked based on how urgently a person needs the money — AI infers financial desperation from behavioral signals.

Banking Data & Surveillance

- Banks know you better than you know yourself — bank transactions tell everything about a person including credit balances and financial history.
- Banking apps detail everything about a user based on spending habits. This data is now being leveraged extensively.
- Home and car loans are the most common forms of secured debt, both used as collateral most frequently.
- Unsecured debt = no collateral attached.
- Snowden revealed that everything is being watched — a reminder that digital surveillance extends well beyond banking.

Anti-Money Laundering (AML)

- Banks use AI to monitor all transactions and ensure transparency as part of Anti-Money Laundering (AML) compliance.

2. Fraud & Anomaly Detection

AI-powered fraud detection is one of the most critical and rapidly evolving areas of financial technology, with real-world cases demonstrating its importance.

Real-World Fraud Case Study

A documented case illustrating modern AI-assisted fraud:

- A 100-year-old victim with a bucket list visited Houston, tried local food, and went out on the town.

- After visiting 5 bars, his phone and wallet were stolen while he was sleeping on a platform at 4am with a woman nearby.
- Cash and ID were intact, but cards were gone.
- The banking app was hacked and converted to Bitcoin — \$75K was gone, transferred out via Robinhood in Bitcoin.
- The entire heist took 1.5 hours. The victim's retina was scanned from the phone and AI was used to bypass login to his accounts — he was wiped clean.

How AI Fraud Works (Malicious Use)

- AI used to detect fraud & login / transactions & location data.
- IP Address Mirroring: emulating IP addresses to steal money — AI fraud is beyond comprehension.
- \$75K was swiped in just 1.5 hours in the documented case.

How AI Detects & Prevents Fraud (Positive Use)

- AI can detect suspicious transactions and flag unusual locations — it can prove if you were actually logged in or not.
- Real-time monitoring allows banks to catch fraud as it happens.
- Chatbots and Agents are now communicating with each other behind the scenes — listening, learning, and remembering everything.
- Banks use AI to build credit & borrowing profiles — shady transactions may flag credit history or prevent credit approval.
- Getting credit is never random — banks' AI systems want to understand the reason behind every application.
- Banks want to avoid fraud because it costs them money — fraud prevention is both protective and profitable.
- Key tools: Credit Risk Scoring & Anomaly Detection.

3. AI in Bank Operations & Customer Experience

Branch Transformation

- Banks are removing drive-thrus, reducing tellers, and operating with only a couple of people in-branch.
- Screens are everywhere — face recognition technology identifies customers by name and occupation upon entry.
- Tellers know your profile, track login history, visit history, and keep a log of all interactions.
- Banks will not dispense money to a person who appears nervous or distressed.

Transaction Monitoring & Security

- AI Detection flags a large number of transactions over \$10K — automatic triggers for review.

- Banks may require identity verification or authentication when suspicious activity is detected.
- Banks also send phishing simulation emails and use AI for phishing prevention and employee training.
- Virtual Arrest Scam Scandal: A notable fraud case originating from India highlighted by the instructor.

Customer Service Automation

- Customer care staff in banks are being replaced by chatbots.
- AI analyzes customer care calls — NLP workloads process conversation data.
- Voice assisting and voice analysis are now integrated into customer support workflows.
- Banks plan cash flows using market demand forecasting powered by AI.

4. AI in Investment & Finance

Market Research & Stock Prediction

- AI agents can help individuals understand finance — even free versions exist, but industrial-grade tools are significantly more powerful.
- Web crawlers parse information from the internet to feed market intelligence to AI systems.
- Agents can be used to find stocks and understand market emotions and sentiment.

Trading Strategies

- Some investors use AI to place automatic calls (options/trades) in the market.
- AI can be used for stock predictions — historical patterns and real-time data inform forecasts.
- Key applications: Preventing Fraud & Optimizing Portfolio management.
- Real estate and home investments can also be optimized using AI.
- Finance optimizations are expanding to serve both businesses and individuals.

5. AI in Insurance & Claims

Fraud Detection & Underwriting

- AI is used for fraud and malpractice detection in insurance claims.
- AI now performs underwriting tasks — evaluating risk automatically based on data.
- AI can initiate claims and process them when given the relevant information.

Claims Processing Automation

- Auto Claim Verification: AI makes calls to public records and automates verification steps.
- AI writes workloads for claims verification — reducing human labor in document processing.
- Invoice processing is automated, reducing errors and increasing speed.

6. Key Takeaways

- Use cases in AI finance are never fully solved — there is always availability to work on and improve them.
- The AI Engineer role is emerging as a critical position in financial services.
- AI is both a tool for consumer benefit and a vector for sophisticated crime — understanding both sides is essential.
- Data is the foundation — bank transactions, credit history, spending habits, and location data are all inputs into modern AI financial systems.